

QUAN TRAN

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EDUCATION

University of Cincinnati

Bachelor of Science in Mechanical Engineering

GPA: 3.8

Cincinnati, Ohio

Expected Graduation: May 2030

EXPERIENCE

Bosch Automotive R&D Center Vietnam – Components & Connector Engineering Department

HCM City, Vietnam

Testing & Validation Engineer Intern

June 2026 – Present

- Executed 5+ validation procedures on 300+ connector samples, including mechanical/electrical, sealing, durability/cycle, and chamber tests according to the USCAR-2, LV214 and JOEM specifications.
- Prepared and loaded 300+ connector samples into thermal, humidity, and thermal shock chambers across 6+ test programs, verifying fixture configuration, chamber conditions, and acceptance criteria per test plan.
- Supported preparation of 40+ pass/fail test reports by documenting test conditions, chamber exposure results, abnormal measurements, and failure behavior to assist engineering review and corrective-action discussions.
- Communicated validation findings and failure modes across testing, design, and project management teams to support component performance evaluation, troubleshooting, and corrective action.

ENGINEERING PROJECTS

RevolutionUC Hackathon – Smart Delivery Box

Cincinnati, Ohio

Mechanical Lead

Feb 2026 – Present

- Led a 4-person team to engineer a laser-cut wood/acrylic enclosure for 30×30×40 cm parcels, delivering parametric CAD/DXF files with ±0.5 mm tolerances.
- Executed validation testing: 600 latch cycles, 60 N retention, 1 m drop at 5 kg, and 200 insertion trials.
- Drove design iterations from test logs to reduce false triggers to <2%.

Hybrid Health Supply Network – Prototype Design & Validation

Cincinnati, Ohio

Project Lead, Mechanical Design

Aug 2025 – Dec 2025

- Directed a 4-member team to define 7 measurable requirements for a drone-enabled medical supply hub concept.
- Built a verification matrix for fit, load, drop, and usability testing, then selected concepts using a weighted decision matrix.
- Fabricated a 3D-printed interface and validated performance with 3-ft drop and 5-lb load testing.

F-Race STEM Competition

HCM City, Vietnam

Mechanical Design Contributor

Sep 2024 – Oct 2024

- Produced AutoCAD part and assembly drawings and maintained revision control for manufacturing clarity.
- Supported technical experts with SolidWorks Simulation/FEA for structural analysis and design improvements.
- Implemented updates that reduced performance issues by 40% and presented results to 15 judges and a 400+ audience.

LEADERSHIP EXPERIENCES

STEM Club – Nguyen Huu Huan High School

HCM City, Vietnam

Head of Technical Division

Sep 2023 – May 2025

- Led 8 technical projects and mentored 40+ members in CAD, including AutoCAD and SolidWorks, supporting concept-to-prototype execution through design reviews and build readiness checks.
- Established repeatable testing and validation workflows, including checklists, equipment setup, pre-run inspections, and issue triage.
- Supervised 6 project teams delivering 15+ prototypes for city-level competitions and coordinated cross-team troubleshooting and technical presentations.

AWARDS & HONORS

• **UC Global Scholarship; International Outreach Award** – University of Cincinnati

Aug 2025

• **STEM Competitions** – Top 8 Water Rocket, City-Level, 2021; Top 4 Water Jet Car, 2022; First Prize Bridge Building, 2023; Innovation Recognition F-Race STEM, 2024.

TECHNICAL SKILLS

Mechanical Design, Simulation & Fabrication: SolidWorks (CSWP), Siemens NX, AutoCAD, GD&T, tolerance analysis, SolidWorks Simulation/FEA, 3D printing, laser cutting, welding, soldering, drilling

Validation & Testing: Automotive connector validation, mechanical/electrical testing, chamber testing, fixture setup, test plan interpretation, pass/fail reporting, failure mode documentation

Programming & Data Analysis: Python, MATLAB, LabVIEW, Microsoft Office (MOS Certified), ESP32 data logging, measurement data analysis

AVAILABLE FOR CO-OP SUMMER 2027